

LLM-driven Ontology Construction for Bioprocesses

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Introduction

The main goal of this project is to develop a multi-scale bioprocess ontology covering micro/mini-bioreactors, lab-scale, pilot-scale, and industrial-scale bioreactors, using at least three Large Language Models (LLMs) as ontology generation assistants. The ontology will provide a shared, machine-interpretable representation of key bioprocess entities and relations (e.g., bioreactor systems, process phases, sensors, actuators, operating variables, events), enabling consistent terminology and knowledge reuse across scales.

First, the project will establish clear ontology requirements by defining the scope and writing a set of competency questions focused on cross-scale bioreactor scenarios. These questions will guide what the ontology must be able to represent and answer in practice, for instance, how to describe which variables are measured or controlled at each scale, how instruments differ between scales, and how equivalent concepts (such as dissolved oxygen measured by different technologies) should be mapped consistently.

Second, the project will apply a comparative LLM-driven workflow. Using a curated bioprocess corpus (e.g., technical documentation, SOPs, variable dictionaries, and relevant scientific literature), three LLMs will be prompted under the same conditions to generate three candidate ontologies, each proposing concepts, taxonomies, and relations. These candidates will then be consolidated into a single OWL ontology through systematic harmonization: deduplication of overlapping concepts, normalization of naming conventions, resolution of synonyms, and alignment of relations and basic constraints (e.g., domain/range where appropriate).

Finally, the ontology will incorporate explicit multi-scale modeling, including bioreactor scale as a first-class concept and representations of sensor/actuator availability and variable equivalences across scales. The resulting ontology will be iteratively validated and refined with domain experts, ensuring conceptual correctness, coherence, and practical utility. The final deliverables will include the expert-validated OWL ontology, brief documentation of scope and modeling decisions, and a change log summarizing the evolution from the LLM-generated candidates to the final version.

Objectives

The main goal of this project is to develop a multi-scale bioprocess ontology (micro/mini, lab, pilot, industrial bioreactors) using at least three LLMs. Specific objectives include:

- Define the ontology scope and a set of competency questions focused on bioreactors across scales.
- Generate three candidate ontologies (one per LLM) from a curated bioprocess corpus.
- Consolidate the candidates into a single OWL ontology, harmonizing concepts, relations, and terminology.
- Include explicit modeling of bioreactor scale, sensor/actuator availability, and variable equivalences across scales.
- Validate and refine the ontology with domain experts, delivering a final version and brief documentation.

Mandatory skills

- Basic knowledge of ontology engineering (OWL/RDF, Protégé, classes/properties/axioms).
- Ability to work with LLMs (prompting, evaluation, reproducibility).
- Python skills for text processing (e.g., parsing corpus, term extraction, normalization).
- Optional: SPARQL, knowledge graphs, semantic integration.

REFERENCES

Giglou, H. B., D'Souza, J., & Auer, S. (2024). *LLMs4OL 2024 Overview: The 1st Large Language Models for Ontology Learning Challenge*. *Open Challenges in Ontology Learning (OCP)*. <https://doi.org/10.52825/ocp.v4i.2473>

Robasky, K., et al. (2026). *MCBO: Mammalian Cell Bioprocessing Ontology, a hub for representing mammalian cell culture bioprocesses in OWL*. *bioRxiv*. <https://doi.org/10.64898/2026.01.05.697007>.